

Illumination Normalisation and Lightweight Attention for Steel Surface Defect Detection: An Audited Empirical Study of a YOLO11s Variant ("IAMA-Net")

Version 3 (2026-10-06). This document supersedes `results/IAMA_Net_Academic_Research_Report.md`. Every numeric table below is generated by `scripts_v3/make_report_tables.py` from evaluation CSVs in `results_v3/` (produced by the scripts named in each caption). The audit of the previous version is `results_v3/audit/audit_report.md`; per-number changes are listed in `results_v3/CHANGELOG.md`.

Abstract

We re-examine "IAMA-Net", a YOLOv11s variant for steel surface defect detection combining (a) CLAHE + bilateral-filter "illumination normalisation" and (b) ECA channel attention with CBAM-style spatial attention at P3/P4/P5, evaluated on NEU-DET with zero-shot class-agnostic transfer to GC10-DET. An external review found that the original study's ablation was confounded (models fine-tuned from a converged baseline with frozen backbones and unequal budgets), its class-wise table came from the wrong checkpoint, and its headline transfer gain rested on 10% of the target dataset. We rebuilt the environment, re-derived every reported number from the original checkpoints, and re-ran the full ablation under a matched protocol (identical COCO-pretrained initialisation, hyperparameters, augmentation, and 100-epoch budget; single seed). Under the matched protocol the picture reverses in part and sharpens in part: the full model improves NEU-DET mAP@0.5 over the baseline by roughly +2 pp (paired image bootstrap $p=0.038$) with mAP@0.5:0.95 inconclusive; the gain is concentrated in the hardest class (crazing, $\approx +10$ pp) and is not specific to the chosen attention modules (SE matches or exceeds them); the preprocessing component shows no measurable in-domain benefit and measurably harms zero-shot transfer; and zero-shot transfer to GC10-DET is near-zero for every variant, with modified models transferring significantly worse than the baseline on all 2,300 images. The previously reported "+49.8% relative transfer gain" and the per-class "regressions" are both shown to be protocol artifacts. All variants run above 55 FPS (batch 1, fp32) on a 4 GB laptop GPU. Results are reported with 95% bootstrap confidence intervals and paired tests; the single-seed limitation is stated explicitly throughout.

1. Introduction

Steel surface defect detection is a standard benchmark problem for industrial vision. The original project asked whether "illumination-aware" preprocessing plus multi-scale lightweight attention improves a YOLO11s detector in-domain (NEU-DET) and under cross-dataset shift (GC10-DET). After an external review identified serious methodological flaws, the contribution was reframed as an **empirical study**:

Does illumination normalisation plus lightweight attention help YOLO-based steel defect detection — in-domain, and under cross-dataset shift — when the comparison is run under a matched protocol with uncertainty quantification?

This document reports (i) a forensic audit of the original claims and checkpoints (Section 4), (ii) the matched-protocol ablation with confidence intervals and paired tests (Section 5.1), (iii) controls for the two interventions — alternative attention modules and corrected

preprocessing variants (Sections 5.2–5.3), (iv) a whole-dataset zero-shot transfer evaluation with explicit confound controls (Section 5.4), and (v) efficiency and quantified interpretability measurements (Sections 5.5–5.6). Negative and inconclusive results are reported as such.

2. Related work

Steel defect detection. NEU-DET (Song & Yan, 2013) [6] and GC10-DET (Lv et al.; see reference note [11]) are the two standard benchmarks. Recent detection work on these datasets is dominated by YOLO-family modifications; e.g. Yasir & Ahn (2023) [9] improve YOLOv5 with depthwise convolutions and channel shuffling on NEU-DET/GC10-DET, and Sun et al. (2025) [8] propose a gamma-correction state-space model evaluated on both.

Attention modules. Squeeze-and-Excitation [3], CBAM [2], ECA [1] and Coordinate Attention [4] are standard lightweight add-ons; ECA achieves channel attention with a 1-D convolution over pooled channel statistics (a handful of parameters). Inserting such modules into YOLO backbones/necks for defect detection is common practice in the 2022–2025 literature [8,9]; this project therefore makes **no novelty claim** for the attention architecture itself.

Illumination/contrast preprocessing. CLAHE [12] and bilateral filtering [7] are classical contrast-enhancement/denoising tools and appear inside many YOLO pipelines for industrial inspection. Section 5.3 evaluates whether they help a modern detector at all — on grayscale input, which makes the original "decouple chromaticity in CIE L*a*b*" rationale inapplicable (both datasets are grayscale; see Section 3.3).

Cross-dataset evaluation. Zero-shot, class-agnostic evaluation on a second plant/dataset is less standardised than in-domain benchmarking; we are not aware of a canonical protocol for NEU-DET→GC10-DET. Our contribution here is methodological: whole-dataset scope, documented class-mapping restrictions, resolution/scale controls, and uncertainty quantification.

3. Method

3.1 Architecture

Baseline: stock YOLOv11s (`configs/yolo11s_baseline.yaml` , `nc=6`). Attention variant ("IAMA"): `ECA_SpatialAttn` blocks inserted after the neck outputs P3/P4/P5 (`configs/yolo11s_iama.yaml` , layers 5/8/13). Each block is ECA (Conv1d(1,1,k), adaptive k) followed by CBAM-style spatial attention (Conv2d(2→1, 7×7) over concat(channel-mean, channel-max), sigmoid gate). Gates are identity-initialised (zero weights, +4.0 bias → cascade scale $\sigma(4)^2 \approx 0.964$).

Recounted from the actual checkpoints (`scripts_v3/count_params.py`):

model	params (unfused)	params (fused)	attention params	ECA kernel sizes
yaml_baseline	9,430,114	9,415,122	0	[]
yaml_iama	9,430,429	9,415,437	315	[5, 5, 5]
ckpt_M1	9,430,114	9,415,122	0	[]
ckpt_M2	9,430,114	9,415,122	0	[]
ckpt_M3	9,430,429	9,415,437	315	[5, 5, 5]
ckpt_M4	9,430,429	9,415,437	315	[5, 5, 5]
ckpt_ANCHOR	9,430,429	9,415,437	315	[5, 5, 5]

Corrections vs the old report: the attention overhead is **315** parameters (the abstract's 309 matches a k=3 assumption; actual ECA kernels are k=5 at all three insertions because the yaml passes unscaled `channels=512` to every block). Actual input channels at P3/P4/P5 are **256/256/512** (not 128/256/512). The old report's baseline count 9,415,122 is the **BN-fused** count; unfused it is 9,430,114.

Reproducibility note. The original training environment patched ultralytics site-packages to register `ECA_SpatialAttn`; that patched class was lost with the venv. The checkpoints store the spatial gate as a raw `nn.Conv2d` named `spatial_conv`, which does not match the repository's `models/attention.py` (wrapper module named `spatial`). We reconstructed a structurally identical class (`scripts_v3/iama_env.py`); the reconstruction is validated by exactly reproducing the originally reported metrics of the affected checkpoints (Section 4).

3.2 Preprocessing variants

- **PP-original ("LAB+bilateral"):** CLAHE (clipLimit 2.0, 8×8 tiles, OpenCV semantics: the clip limit is a multiplier of the mean histogram-bin height) on the L* channel of a BGR→LAB round trip, then `bilateralFilter(d=5, σ_color=σ_space=50)`. Parameter identity was verified by re-applying candidates to raw images and comparing against the stored cache (`results_v3/audit/preprocessing_param_id.csv`: MAE 1.47 vs ≥4.79 for all alternatives — residual error is JPEG re-encoding noise).
- **PP-gray (control, Section 5.3):** CLAHE (clip 2.0, 8×8) directly on the single grayscale channel; no LAB round trip; no bilateral filter.

Both datasets are grayscale. The LAB round trip is therefore not motivated by chromaticity; it is also measurably lossy: after conversion, 308 of 1,797 preprocessed NEU images no longer have identical BGR channels (chroma noise introduced by the round trip + JPEG coding; `results_v3/audit/data_audit_summary.json`).

3.3 Datasets

dataset	split	images	boxes	size (px)	channels
NEU-DET	train	1257	2918	200x200	3ch-identical(grayscale-stored-as-BGR)
NEU-DET	val	270	647	200x200	3ch-identical(grayscale-stored-as-BGR)
NEU-DET	test	270	612	200x200	3ch-identical(grayscale-stored-as-BGR)
NEU-DET_preprocessed	train	1257	2918	200x200	3ch-identical(grayscale-stored-as-BGR), 3ch-color
NEU-DET_preprocessed	val	270	647	200x200	3ch-identical(grayscale-stored-as-BGR), 3ch-color
NEU-DET_preprocessed	test	270	612	200x200	3ch-identical(grayscale-stored-as-BGR), 3ch-color
GC10-DET	train	1840	2848	2048x1000	1ch
GC10-DET	val	230	349	2048x1000	1ch
GC10-DET	test	230	366	2048x1000	1ch

GC10-DET box counts cross-checked against the dataset's own `metadata.jsonl` annotations:

scope	boxes
total	3563
train	2848
val	349
test	366
per class	crease=74; crescent_gap=265; inclusion=347; oil_spot=569; punching_hole=329; rolled_pit=85; silk_spot=884; waist_folding=143; water_spot=354; welding_line=513

Corrections vs the old report: GC10-DET locally comprises **2,300 images** (the "2,303" counted three `metadata.jsonl` files) with **3,563 boxes**; its 10th class is `rolled_pit`, not "rolled-in scale"; NEU-DET images are 200×200 (not "high-resolution"). Split integrity was audited (SHA-256 of file bytes and decoded pixels): **no duplicates and no train/val/test leakage** in either dataset (`scripts_v3/audit_data.py`).

3.4 Training protocols

Old (confounded) protocol — recorded from `runs/*/args.yaml` :

run	epochs	init	optimiser / lr0	cos LR	warmup	close_mosaic	frozen layers	data
M1	100	yaml + COCO transfer	auto / 0.01	no	3.0	10	none	raw
M2	25	M1 best.pt	AdamW / 0.0008	yes	1.0	0	none	preprocessed
M3	25	identity-mapped M1 best	AdamW / 0.0008	yes	1.0	0	10 backbone layers	raw
M4	30	identity-mapped M1 best	AdamW / 0.0008	yes	1.0	0	10 backbone layers	preprocessed

The old report's claim that all models were trained "under identical hyperparameters" is false (see audit §9). M2–M4 are short fine-tunes of the converged M1 — with a frozen backbone for the attention variants — so their deltas conflate the interventions with fine-tuning dynamics. Evidence that this mattered: the

identity-mapped anchor

(M1 weights ported into the attention architecture with pass-through gates, i.e. functionally $\approx$ M1) scored **higher** than every trained variant on the NEU test split (0.7608 vs 0.7527/0.7444/0.7484/0.7333).

Matched protocol (this study) — `scripts_v3/train_matched.py`, one run per variant, seed 0, 100 epochs, all variants from the same COCO-pretrained YOLO11s (`yolo11s.pt` ; attention variants via aligned semantic layer mapping, 493/499 tensors transferred — the 6 skipped are the nc-mismatched Detect class convolutions; manifest: `runs_v3/init/iama_coco_mapped_init.manifest.json` ; identity attention init preserved). AdamW lr0=8e-4, lrf=0.01, wd=5e-4, cosine schedule, warmup 1 epoch, close_mosaic=0, batch 8 (accumulate 2 via nbs=16), imgsz 640, workers 2, deterministic=True, no freezing, identical augmentation (ultralytics defaults, pinned in the manifest). Hardware: RTX 3050 Laptop 4 GB, Windows 11, torch 2.11.0+cu128, ultralytics 8.3.40. Per-run manifests (command, seed, git commit, epochs, timings): `runs_v3/*_matched_seed0/manifest.json` . Measured cost $\approx$ 62–87 min per 100-epoch run.

Variants: **M1** baseline/raw, **M2** baseline/PP-original, **M3** IAMA/raw, **M4** IAMA/PP-original (Phase 2); **M3se**, **M3cbam** (SE / CBAM at the same three insertion points, same identity-init principle, raw inputs) and **M2gray** (baseline arch, PP-gray) as Phase 3 controls.

Limitation (user-directed): a single seed per variant. Between-seed variance cannot be estimated; all uncertainty statements are image-level bootstrap CIs conditional on these runs.

3.5 Evaluation protocols

All evaluation uses ultralytics `val()` semantics (rect letterboxing, conf=0.001, NMS IoU=0.7, imgsz=640, max_det=300), each model evaluated on the input distribution it was trained on, plus explicit cross-distribution diagnostics. Per-image detections are cached (`val(save_json=True)` → JSONL) and re-scored by an independent COCO-style implementation (`scripts_v3/metrics_lib.py`) that reproduces ultralytics mAP exactly ($\Delta \leq 0.0001$ on NEU).

- **NEU-DET:** test split (270 images, 612 boxes); macro mAP@0.5, mAP@0.5:0.95, per-class AP, mean P/R at the max-mean-F1 confidence.
- **GC10-DET zero-shot:** class-agnostic (single_cls: all predictions merged, cross-class NMS, all GT collapsed), on **all 2,300 images** (primary) and the 230-image test split (for reconciliation with the old protocol).
- **Restricted-overlap protocol:** predictions restricted to NEU {inclusion, pitted_surface, rolled-in_scale} vs GT restricted to GC10 {inclusion, rolled_pit}, collapsed to one class (mapping fixed before evaluation and justified in `scripts_v3/gc10_restricted_overlap.py`).
- **Uncertainty:** 95% percentile CIs from 1,000 image-level bootstrap resamples (seed 20261005); paired bootstrap (same resampled images for both models) for variant-vs-baseline deltas, two-sided $p = 2 \cdot \min(P(\Delta \leq 0), P(\Delta \geq 0))$. "Significant" is used only where the paired CI excludes 0.
- **FPS:** batch 1, 50 warm-up + 300 timed iterations × 5 repeats, fp32/fp16 separately, real test image and random-noise input, preprocessing/inference/NMS split, GPU clocks and temperature logged (`scripts_v3/fps_bench.py`, `gpu_state.csv` files).
- **Grad-CAM quantification:** gradient-weighted CAM (target = highest pre-NMS anchor class score) at P3/P4/P5 and fused, on the whole NEU test split; pointing game (argmax inside any GT box; unconditional and conditional on a TP detection at conf ≥ 0.25) and energy-in-GT-box, with image-bootstrap CIs (`scripts_v3/gradcam_quant.py`).

4. Audit of the original report

Full detail: `results_v3/audit/audit_report.md` (generator: `scripts_v3/make_audit_report.py`).

Summary of findings:

1. **Master-table mAP@0.5 values are genuine:** all reproduce within ± 0.0002 from the original checkpoints under a fixed protocol; batch-size sensitivity is ≤ 0.0002 (`results_v3/audit/val_batch_sensitivity.csv`).
2. **The class-wise table was computed from the wrong checkpoint:** the report's "M1" column matches the identity-mapped anchor (total |deviation| 0.0019) and not M1 (0.0646). The M2/M3/M4 class columns do not match their own checkpoints either (max per-class deviation 0.10–0.14). The headline claim "inclusion 78.1 → 87.8 (+9.7 pp)" is **not reproducible from any checkpoint in the repository**: recomputed inclusion AP50 spans 0.782–0.788 across all five original models.

model	metric	old reported	new val()	$\Delta(\text{new}-\text{old})$
M1	mAP@0.5	0.7527	0.7527	0.0000
M1	mAP@0.5:0.95	—	0.4202	—
M1	precision	0.7101	0.7099	-0.0002
M1	recall	0.6898	0.6896	-0.0002
M2	mAP@0.5	0.7444	0.7444	0.0000
M2	mAP@0.5:0.95	—	0.4275	—
M2	precision	0.6912	0.6910	-0.0002
M2	recall	0.6924	0.6924	-0.0000
M3	mAP@0.5	0.7484	0.7484	0.0000
M3	mAP@0.5:0.95	—	0.4254	—
M3	precision	0.6721	0.6729	0.0008
M3	recall	0.7082	0.7082	-0.0000
M4	mAP@0.5	0.7333	0.7333	-0.0000
M4	mAP@0.5:0.95	—	0.4058	—
M4	precision	0.6686	0.6704	0.0018
M4	recall	0.6815	0.6835	0.0020
ANCHOR	mAP@0.5	0.7608	0.7607	-0.0001
ANCHOR	mAP@0.5:0.95	—	0.4185	—
ANCHOR	precision	0.7114	0.7115	0.0001
ANCHOR	recall	0.6907	0.6907	0.0000

class	rep M1	new M1	new ANCHOR	rep M2	new M2	rep M3	new M3	rep M4	new M4
crazing	0.49	0.449	0.489	0.412	0.393	0.434	0.420	0.386	0.397
inclusion	0.781	0.782	0.781	0.866	0.787	0.878	0.788	0.865	0.784
patches	0.944	0.946	0.944	0.791	0.926	0.813	0.943	0.812	0.915
pitted_surface	0.803	0.808	0.803	0.814	0.811	0.831	0.839	0.813	0.796
rolled-in_scale	0.658	0.643	0.658	0.602	0.666	0.583	0.640	0.569	0.631
scratches	0.889	0.888	0.889	0.918	0.884	0.926	0.860	0.931	0.876
MEAN	0.7608	0.753	0.761	0.7338	0.744	0.7442	0.748	0.7293	0.733

- mAP@0.5:0.95 column has no CSV provenance** in the repo and does not reproduce (e.g. M2: reported 0.4070 vs recomputed 0.4275).
- Parameter overhead is 315, not 309**; ECA kernels are k=5; P3/P4 are 256-channel (Section 3.1).
- Dataset facts corrected** (Section 3.3), including rolled_pit as GC10's 10th class and 2,300/3,563 image/box counts; the LAB route injects chroma noise into grayscale data.

6. **FPS numbers are not reproducible** under a controlled protocol; the old 43.5–59.5 spread between variants overstates architecture costs (Section 5.5).
7. **The ablation was confounded** (Section 3.4), and the "+49.8% transfer gain" relied on the 230-image test split: on all 2,300 zero-shot images the original M4–M1 difference is +0.32 pp with a CI spanning zero ($p=0.17$; `results_v3/audit/paired_bootstrap_m4_vs_m1.csv`).
Original-model re-evaluation with CIs:

model	dataset	inputs	scope	metric	point	95% CI
M1	neu	raw	all	mAP@0.5	0.7527	[0.7210, 0.7879]
M1	neu	raw	all	mAP@0.5:0.95	0.4202	[0.3962, 0.4553]
M2	neu	pp	all	mAP@0.5	0.7444	[0.7152, 0.7826]
M2	neu	pp	all	mAP@0.5:0.95	0.4275	[0.4047, 0.4616]
M3	neu	raw	all	mAP@0.5	0.7484	[0.7167, 0.7855]
M3	neu	raw	all	mAP@0.5:0.95	0.4254	[0.4013, 0.4603]
M4	neu	pp	all	mAP@0.5	0.7333	[0.7032, 0.7712]
M4	neu	pp	all	mAP@0.5:0.95	0.4058	[0.3824, 0.4394]
ANCHOR	neu	raw	all	mAP@0.5	0.7607	[0.7279, 0.7960]
ANCHOR	neu	raw	all	mAP@0.5:0.95	0.4185	[0.3962, 0.4539]
M2	neu	raw	all	mAP@0.5	0.5407	[0.5099, 0.5819]
M2	neu	raw	all	mAP@0.5:0.95	0.2875	[0.2697, 0.3182]
M4	neu	raw	all	mAP@0.5	0.5563	[0.5264, 0.5939]
M4	neu	raw	all	mAP@0.5:0.95	0.2949	[0.2755, 0.3257]
M1	gc10	raw	all	mAP@0.5	0.0380	[0.0343, 0.0439]
M1	gc10	raw	all	mAP@0.5:0.95	0.0124	[0.0108, 0.0149]
M2	gc10	raw	all	mAP@0.5	0.0312	[0.0253, 0.0345]
M2	gc10	raw	all	mAP@0.5:0.95	0.0101	[0.0075, 0.0115]
M3	gc10	raw	all	mAP@0.5	0.0382	[0.0320, 0.0461]
M3	gc10	raw	all	mAP@0.5:0.95	0.0113	[0.0097, 0.0139]
M4	gc10	raw	all	mAP@0.5	0.0413	[0.0361, 0.0501]
M4	gc10	raw	all	mAP@0.5:0.95	0.0123	[0.0101, 0.0143]
ANCHOR	gc10	raw	all	mAP@0.5	0.0402	[0.0361, 0.0455]
ANCHOR	gc10	raw	all	mAP@0.5:0.95	0.0135	[0.0115, 0.0158]
M2	gc10	pp	all	mAP@0.5	0.0373	[0.0333, 0.0431]
M2	gc10	pp	all	mAP@0.5:0.95	0.0118	[0.0102, 0.0139]
M4	gc10	pp	all	mAP@0.5	0.0449	[0.0395, 0.0540]
M4	gc10	pp	all	mAP@0.5:0.95	0.0139	[0.0120, 0.0165]
M1	gc10	raw	test	mAP@0.5	0.0313	[0.0200, 0.0498]
M1	gc10	raw	test	mAP@0.5:0.95	0.0096	[0.0057, 0.0165]
M2	gc10	raw	test	mAP@0.5	0.0282	[0.0202, 0.0413]
M2	gc10	raw	test	mAP@0.5:0.95	0.0091	[0.0063, 0.0142]

model	dataset	inputs	scope	metric	point	95% CI
M3	gc10	raw	test	mAP@0.5	0.0284	[0.0180, 0.0485]
M3	gc10	raw	test	mAP@0.5:0.95	0.0100	[0.0047, 0.0175]
M4	gc10	raw	test	mAP@0.5	0.0458	[0.0289, 0.0719]
M4	gc10	raw	test	mAP@0.5:0.95	0.0138	[0.0079, 0.0246]
ANCHOR	gc10	raw	test	mAP@0.5	0.0327	[0.0229, 0.0497]
ANCHOR	gc10	raw	test	mAP@0.5:0.95	0.0099	[0.0063, 0.0168]

comparison	metric	M1	M4	Δ	Δ 95% CI	p (two-sided)
NEU-DET test (M1 raw vs M4 preprocessed inputs)	mAP@0.5	0.7527	0.7333	-1.94 pp	[-3.90, 0.17] pp	0.0720
NEU-DET test (M1 raw vs M4 preprocessed inputs)	mAP@0.5:0.95	0.4202	0.4058	-1.44 pp	[-2.90, -0.17] pp	0.0280
GC10 all images, raw inputs	mAP@0.5	0.0380	0.0413	0.32 pp	[-0.11, 0.89] pp	0.1700
GC10 all images, raw inputs	mAP@0.5:0.95	0.0124	0.0123	-0.01 pp	[-0.28, 0.15] pp	0.7020
GC10 test split, raw inputs	mAP@0.5	0.0313	0.0458	1.45 pp	[0.43, 3.15] pp	0.0140
GC10 test split, raw inputs	mAP@0.5:0.95	0.0096	0.0138	0.42 pp	[0.03, 1.11] pp	0.0320

model	inputs	scope	n GT (mapped)	mAP@0.5	95% CI
M1	raw	all	432	0.00195	[0.00133, 0.00318]
M2	raw	all	432	0.00074	[0.00052, 0.00103]
M3	raw	all	432	0.00153	[0.00106, 0.00244]
M4	raw	all	432	0.00110	[0.00070, 0.00189]
ANCHOR	raw	all	432	0.00203	[0.00144, 0.00313]
M2	pp	all	432	0.00104	[0.00073, 0.00152]
M4	pp	all	432	0.00220	[0.00143, 0.00337]
M1	raw	test	31	0.00086	[0.00025, 0.00234]
M2	raw	test	31	0.00040	[0.00007, 0.00160]
M3	raw	test	31	0.00042	[0.00006, 0.00110]
M4	raw	test	31	0.00059	[0.00011, 0.00303]
ANCHOR	raw	test	31	0.00108	[0.00023, 0.00273]

5. Results (matched protocol, seed 0, 100 epochs)

5.1 In-domain ablation (NEU-DET test)

Generator: `scripts_v3/eval_unified.py` + `scripts_v3/bootstrap_ci.py` +
`scripts_v3/aggregate_matched.py` → `results_v3/ablation_matched.csv`.

M2/M4 trained and evaluated on PP-original inputs; M1/M3 on raw.

variant	arch	train inputs	mAP@0.5	95% CI	mAP@0.5:0.95	95% CI	P	R	Δ mAP50 vs M1 (pp)	p
M1	baseline	raw	0.7345	[0.704, 0.768]	0.4212	[0.398, 0.455]	0.657	0.703		
M2	baseline	preprocessed	0.7376	[0.709, 0.776]	0.4190	[0.393, 0.455]	0.694	0.674	0.3113	0.668
M3	iama_attention	raw	0.7460	[0.714, 0.780]	0.4237	[0.401, 0.457]	0.705	0.688	1.157	0.178
M4	iama_attention	preprocessed	0.7547	[0.726, 0.790]	0.4258	[0.403, 0.458]	0.674	0.700	2.0235	0.038

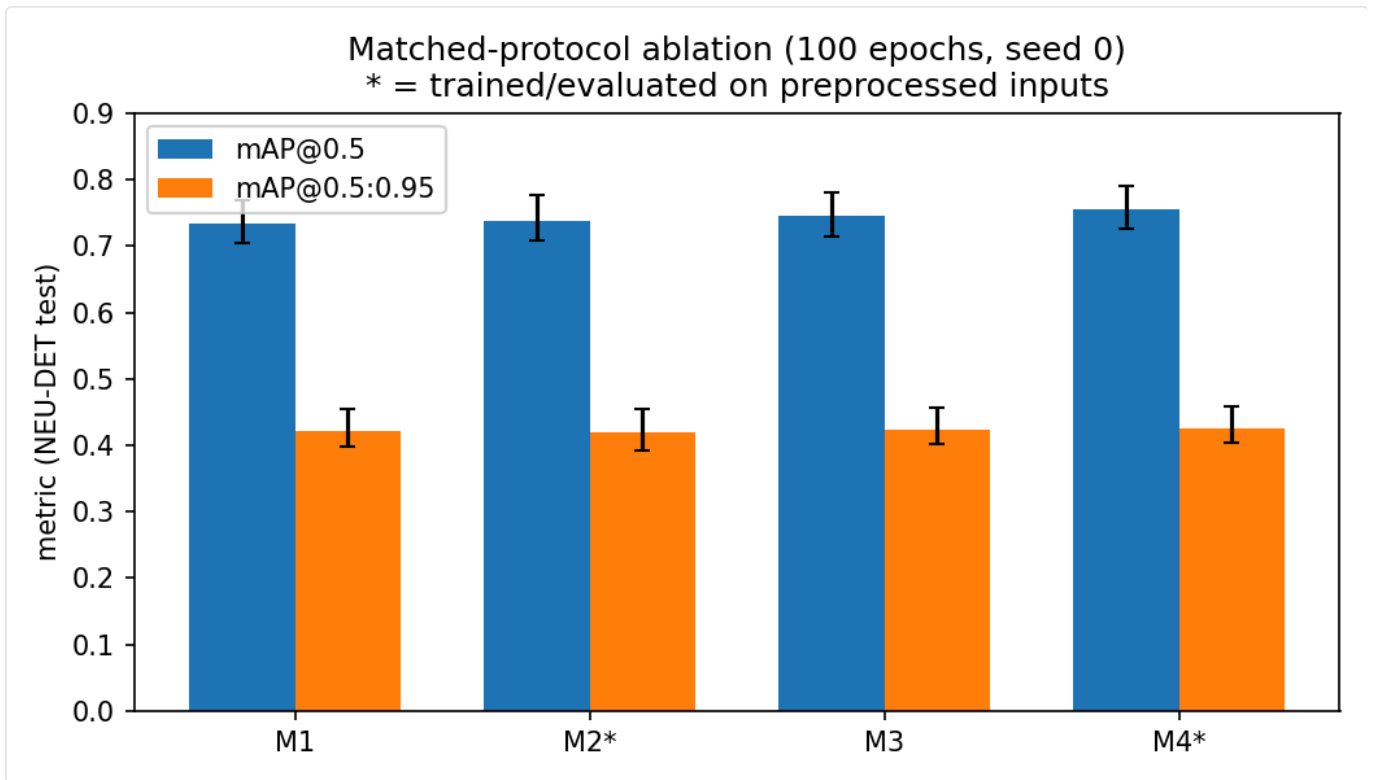


Figure: Matched-protocol ablation on NEU-DET test with 95% image-bootstrap CIs.

Per-class AP@0.5:

variant	crazing	inclusion	patches	pitted_surface	rolled-in_scale	scratches
M1	0.346	0.772	0.942	0.839	0.612	0.896
M2	0.357	0.781	0.951	0.809	0.636	0.891
M3	0.431	0.757	0.940	0.791	0.659	0.900
M4	0.450	0.784	0.938	0.804	0.637	0.914

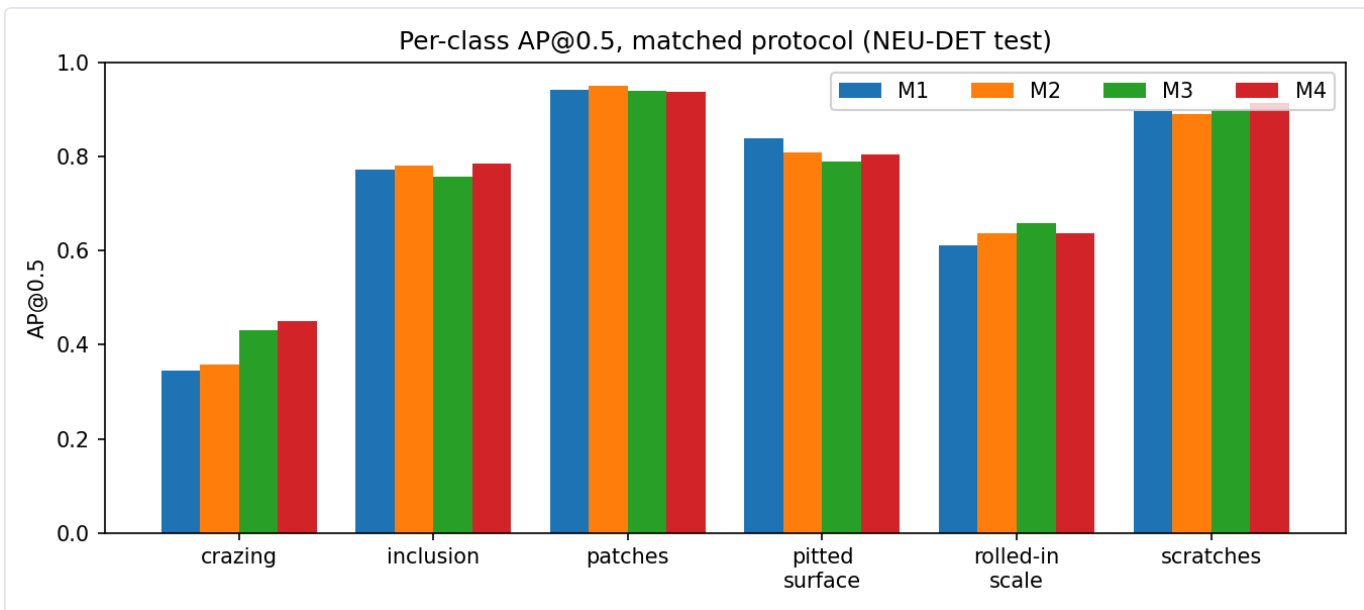


Figure: Per-class AP@0.5 under the matched protocol.

Findings. (i) The full model M4 is the best variant in-domain: +2.02 pp mAP@0.5 over baseline (paired bootstrap $p=0.038$; CI excludes 0), while mAP@0.5:0.95 (+0.46 pp, $p=0.41$) is inconclusive. (ii) The gain is class-concentrated: crazing +10.5 pp (0.346 $\rightarrow$ 0.450), scratches +1.8 pp, rolled-in scale +2.5 pp; pitted_surface -3.5 pp is the only consistent negative (within wide CIs; see 5.2 and Limitations). (iii) Preprocessing alone changes nothing measurably (+0.31 pp, $p=0.67$; mAP@0.5:0.95 -0.22 pp, $p=0.72$). (iv) Attention alone is suggestive but not significant on its own (+1.16 pp, $p=0.18$). (v) The old report's per-class "regressions" (crazing -10.4 pp, patches -13.2 pp, rolled-in scale -8.9 pp for M4) **reverse sign or vanish** under the matched protocol — they were artifacts of fine-tuning a converged model with a frozen backbone, not effects of preprocessing or attention.

5.2 Alternative attention modules (control)

Generator: `scripts_v3/aggregate_phase3.py` $\rightarrow$ `results_v3/attention_alternatives.csv`.
SE and CBAM at the same insertion points, same identity-init principle, same protocol; all attention-only (raw inputs).

attention	id	params	NEU mAP@0.5	95% CI	NEU mAP@0.5:0.95	GC10-all mAP@0.5	Δ NEU vs ECA+spatial (pp)	p	FPS fp32
none (baseline)	M1m	9,430,114	0.7345	[0.70361, 0.768362]	0.4212	0.0448			65.03
ECA+spatial (IAMA)	M3m	9,430,429	0.7460	[0.714455, 0.779948]	0.4237	0.0283			59.92
SE	M3se	9,480,354	0.7491	[0.718196, 0.785586]	0.4238	0.0323	0.3019	0.798	66.64
CBAM	M3cbam	9,480,651	0.7408	[0.709532, 0.779352]	0.4224	0.0287	-0.5225	0.578	62.07

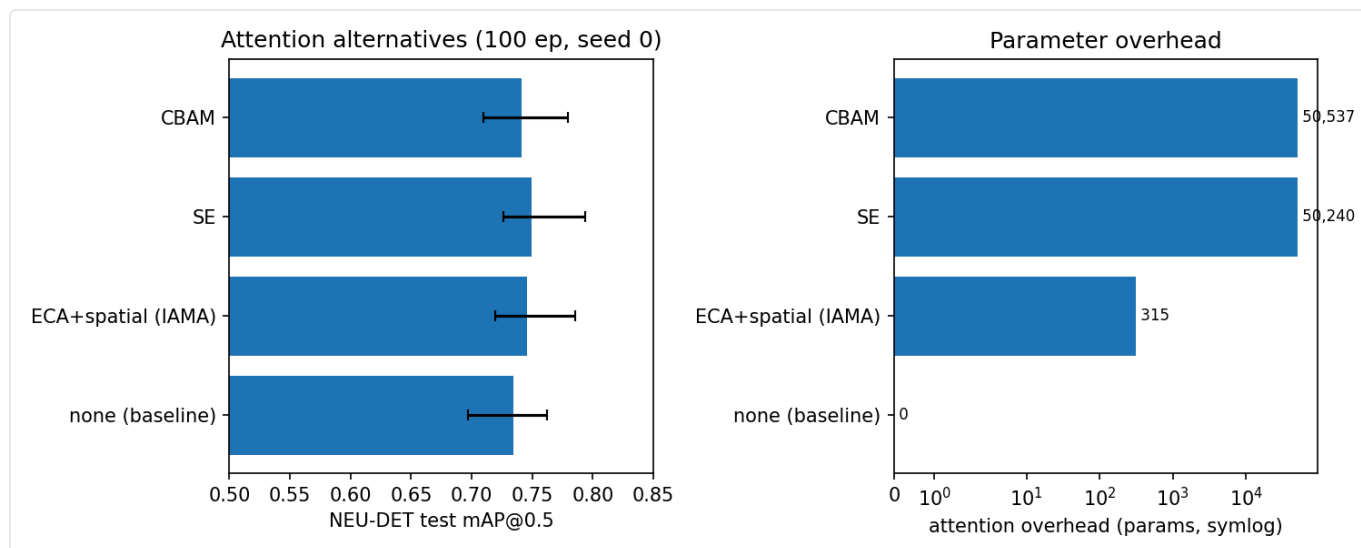


Figure: Attention alternatives: accuracy vs parameter overhead.

Findings. SE matches ECA+spatial in-domain (+0.30 pp, $p=0.80$) and is significantly better on GC10 all-images zero-shot (+0.41 pp, $p=0.006$); CBAM is statistically indistinguishable from ECA+spatial in-domain (-0.52 pp, $p=0.58$). All attention variants improve crazing by +8.5 to +10.6 pp over baseline. Conclusion: the in-domain benefit is attributable to adding attention

at P3/P4/P5, not to the specific ECA+spatial design; on the transfer criterion SE dominates. The pitted_surface drop appears for ECA+spatial and SE (-3.5 to -4.9 pp vs baseline) but **not** for CBAM (-0.3 pp), and also for both preprocessing variants (-3.0 to -4.1 pp) — with only 67 test instances (CI half-width $\approx \pm 9$ pp) and one seed, it cannot be attributed to a single factor.

5.3 Preprocessing controls

Generator: `scripts_v3/aggregate_phase3.py` → `results_v3/preprocessing_controls.csv`.

PP-gray = grayscale CLAHE, no LAB round trip, no bilateral filter.

preprocessing	id	NEU mAP@0.5	95% CI	NEU mAP@0.5:0.95	AP50 crazing	Δ vs LAB+bilateral (pp)	p
none (raw)	M1m	0.7345	[0.70361, 0.768362]	0.4212	0.346		
LAB CLAHE + bilateral d5 s50 (original)	M2m	0.7376	[0.708972, 0.776292]	0.4190	0.357		
grayscale CLAHE, no bilateral	M2gray	0.7257	[0.695105, 0.761312]	0.4072	0.389	-1.19	0.256

Findings. No preprocessing variant shows an in-domain benefit over raw inputs; the "principled" grayscale CLAHE control is 1.19 pp

below

PP-original ($p=0.26$, CI spans 0) and significantly below it on GC10 zero-shot (-0.97 pp, $p<0.001$). Whatever small in-domain difference PP-original makes is within noise; its transfer cost is not. The original justification (decoupling chromaticity in LAB) is inapplicable to grayscale data, and the LAB

round trip measurably injects chroma noise (Section 3.2). We therefore report H1 (preprocessing helps) as **not supported**.

5.4 Zero-shot cross-dataset transfer (GC10-DET)

Generators: `scripts_v3/eval_unified.py`, `scripts_v3/bootstrap_ci.py`,
`scripts_v3/gc10_restricted_overlap.py`, `scripts_v3/transfer_controls.py` →
`results_v3/cross_dataset_matched.csv`, `results_v3/transfer_controls.csv` .

variant	protocol	mAP@0.5	95% CI	P	R	Δ vs M1 (pp)	p
M1	agnostic_test230_raw	0.0419		0.1365	0.1585		
M1	agnostic_all2300_raw	0.0448	[0.0403, 0.0510]	0.1327	0.1527		
M1	agnostic_all2300_raw_testsubset	0.0448	[0.0297, 0.0647]	0.1327	0.1527		
M1	restricted_all_raw	—	[0.0011, 0.0026]	—	—		
M1	restricted_test_raw	—	[0.0004, 0.0025]	—	—		
M2	agnostic_test230_raw	0.0516		0.1121	0.1204		
M2	agnostic_all2300_raw	0.0370	[0.0342, 0.0440]	0.1010	0.1257		
M2	agnostic_all2300_raw_testsubset	0.0370	[0.0352, 0.0748]	0.1010	0.1257		
M2	agnostic_all2300_pp	0.0286	[0.0260, 0.0317]	0.0713	0.1089		
M2	agnostic_test230_pp	0.0287		0.0771	0.0902		
M2	restricted_all_raw	—	[0.0006, 0.0011]	—	—		
M2	restricted_all_pp	—	[0.0004, 0.0008]	—	—		
M2	restricted_test_raw	—	[0.0001, 0.0014]	—	—		
M2	paired_delta_vs_M1_all2300	—		—	—	-0.7287	0.014
M2	paired_delta_vs_M1_test230	—		—	—	0.982	0.402
M3	agnostic_test230_raw	0.0289		0.0771	0.1093		
M3	agnostic_all2300_raw	0.0283	[0.0255, 0.0318]	0.0810	0.1128		
M3	agnostic_all2300_raw_testsubset	0.0283	[0.0185, 0.0429]	0.0810	0.1128		
M3	restricted_all_raw	—	[0.0011, 0.0025]	—	—		
M3	restricted_test_raw	—	[0.0004, 0.0020]	—	—		
M3	paired_delta_vs_M1_all2300	—		—	—	-1.6163	0.0
M3	paired_delta_vs_M1_test230	—		—	—	-1.2962	0.0
M4	agnostic_test230_raw	0.0415		0.1122	0.1093		
M4	agnostic_all2300_raw	0.0339	[0.0314, 0.0380]	0.1073	0.1114		
M4	agnostic_all2300_raw_testsubset	0.0339	[0.0284, 0.0619]	0.1073	0.1114		
M4	agnostic_all2300_pp	0.0245	[0.0226, 0.0277]	0.0680	0.1083		
M4	agnostic_test230_pp	0.0224		0.0752	0.1148		
M4	restricted_all_raw	—	[0.0003, 0.0007]	—	—		
M4	restricted_all_pp	—	[0.0002, 0.0005]	—	—		
M4	restricted_test_raw	—	[0.0000, 0.0008]	—	—		
M4	paired_delta_vs_M1_all2300	—		—	—	-1.0332	0.0
M4	paired_delta_vs_M1_test230	—		—	—	0.0628	0.792

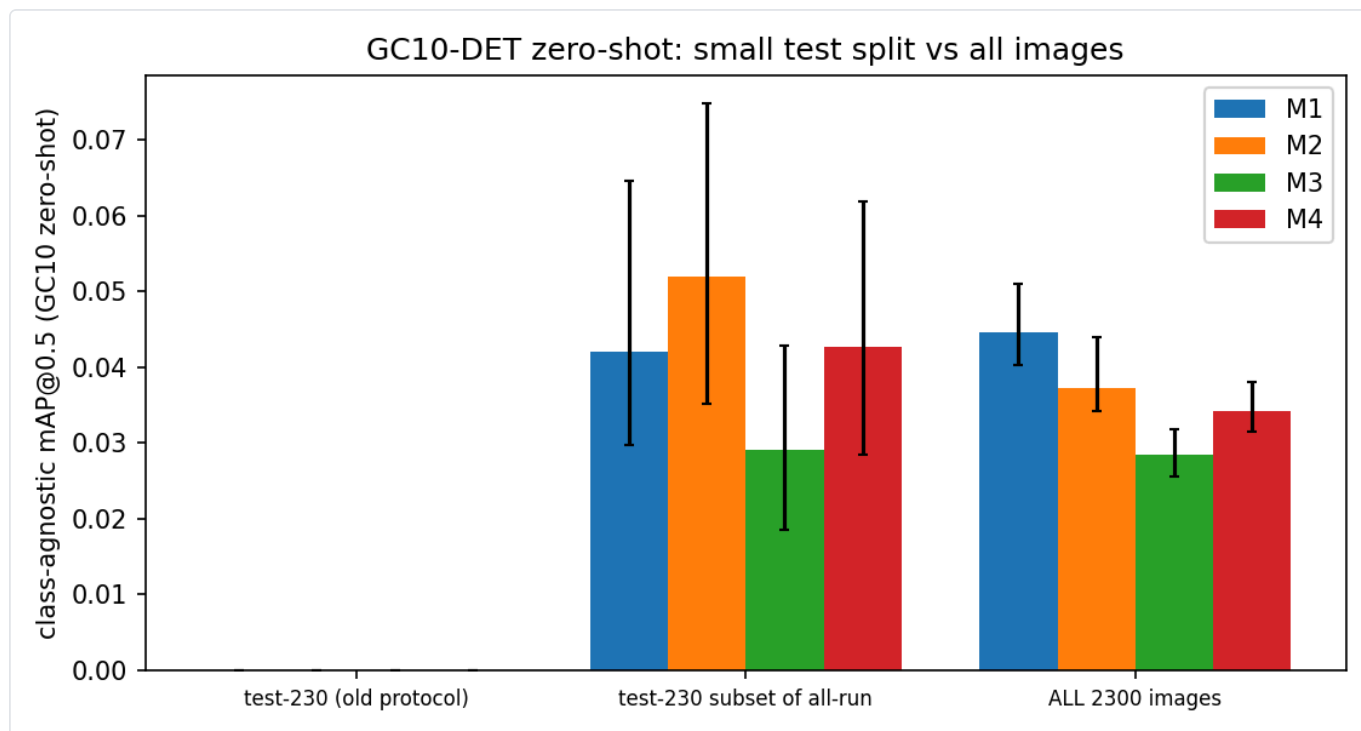


Figure: GC10 zero-shot mAP@0.5: old 230-image protocol vs all 2,300 images.

Confounding controls (matched models):

control	model	setting	metric	value	95% CI	R@0.25
multiscale_agnostic_all2300	M1m	imgsz=640	mAP@0.5	0.044836		
multiscale_agnostic_all2300	M1m	imgsz=1024	mAP@0.5	0.064445		
multiscale_agnostic_all2300	M1m	imgsz=1280	mAP@0.5	0.055427		
multiscale_agnostic_all2300	M4m	imgsz=640	mAP@0.5	0.033872		
multiscale_agnostic_all2300	M4m	imgsz=1024	mAP@0.5	0.038997		
multiscale_agnostic_all2300	M4m	imgsz=1280	mAP@0.5	0.035897		
tiling_640_defect_scale_matched	M1m	tiled_640_stride640_agnostic_nms0.7	mAP@0.5	0.024393	[0.020103, 0.032142]	
tiling_640_defect_scale_matched	M4m	tiled_640_stride640_agnostic_nms0.7	mAP@0.5	0.014997	[0.009489, 0.016753]	
per_gc10_class_agnostic	M1m	crease	AP50 + recall@0.25	0.000155		0.027027
per_gc10_class_agnostic	M1m	crescent_gap	AP50 + recall@0.25	0.01661		0.279245
per_gc10_class_agnostic	M1m	inclusion	AP50 + recall@0.25	0.001069		0.017291
per_gc10_class_agnostic	M1m	oil_spot	AP50 + recall@0.25	0.042799		0.161687
per_gc10_class_agnostic	M1m	punching_hole	AP50 + recall@0.25	0.036045		0.382979
per_gc10_class_agnostic	M1m	rolled_pit	AP50 + recall@0.25	0.00085		0.094118
per_gc10_class_agnostic	M1m	silk_spot	AP50 + recall@0.25	0.001531		0.002262
per_gc10_class_agnostic	M1m	waist_folding	AP50 + recall@0.25	0.0		0.0
per_gc10_class_agnostic	M1m	water_spot	AP50 + recall@0.25	0.010895		0.161017
per_gc10_class_agnostic	M1m	welding_line	AP50 + recall@0.25	6e-06		0.0
per_gc10_class_agnostic	M2m	crease	AP50 + recall@0.25	0.000131		0.013514
per_gc10_class_agnostic	M2m	crescent_gap	AP50 + recall@0.25	0.042968		0.30566
per_gc10_class_agnostic	M2m	inclusion	AP50 + recall@0.25	0.000379		0.008646
per_gc10_class_agnostic	M2m	oil_spot	AP50 + recall@0.25	0.01753		0.084359
per_gc10_class_agnostic	M2m	punching_hole	AP50 + recall@0.25	0.010608		0.018237

control	model	setting	metric	value	95% CI	R@0.25
per_gc10_class_agnostic	M2m	rolled_pit	AP50 + recall@0.25	0.00018		0.0
per_gc10_class_agnostic	M2m	silk_spot	AP50 + recall@0.25	0.008981		0.05543
per_gc10_class_agnostic	M2m	waist_folding	AP50 + recall@0.25	8e-05		0.0
per_gc10_class_agnostic	M2m	water_spot	AP50 + recall@0.25	0.004249		0.025424
per_gc10_class_agnostic	M2m	welding_line	AP50 + recall@0.25	2.2e-05		0.0
per_gc10_class_agnostic	M3m	crease	AP50 + recall@0.25	0.000348		0.067568
per_gc10_class_agnostic	M3m	crescent_gap	AP50 + recall@0.25	0.014043		0.120755
per_gc10_class_agnostic	M3m	inclusion	AP50 + recall@0.25	0.000892		0.023055
per_gc10_class_agnostic	M3m	oil_spot	AP50 + recall@0.25	0.031763		0.147627
per_gc10_class_agnostic	M3m	punching_hole	AP50 + recall@0.25	0.008449		0.030395
per_gc10_class_agnostic	M3m	rolled_pit	AP50 + recall@0.25	0.000767		0.023529
per_gc10_class_agnostic	M3m	silk_spot	AP50 + recall@0.25	0.000291		0.001131
per_gc10_class_agnostic	M3m	waist_folding	AP50 + recall@0.25	0.0		0.0
per_gc10_class_agnostic	M3m	water_spot	AP50 + recall@0.25	0.007144		0.053672
per_gc10_class_agnostic	M3m	welding_line	AP50 + recall@0.25	3.7e-05		0.0
per_gc10_class_agnostic	M4m	crease	AP50 + recall@0.25	8.2e-05		0.0
per_gc10_class_agnostic	M4m	crescent_gap	AP50 + recall@0.25	0.028908		0.184906
per_gc10_class_agnostic	M4m	inclusion	AP50 + recall@0.25	0.000256		0.002882
per_gc10_class_agnostic	M4m	oil_spot	AP50 + recall@0.25	0.013935		0.068541
per_gc10_class_agnostic	M4m	punching_hole	AP50 + recall@0.25	0.01731		0.018237

control	model	setting	metric	value	95% CI	R@0.25
per_gc10_class_agnostic	M4m	rolled_pit	AP50 + recall@0.25	4.7e-05		0.0
per_gc10_class_agnostic	M4m	silk_spot	AP50 + recall@0.25	0.011604		0.039593
per_gc10_class_agnostic	M4m	waist_folding	AP50 + recall@0.25	0.000347		0.006993
per_gc10_class_agnostic	M4m	water_spot	AP50 + recall@0.25	0.002085		0.011299
per_gc10_class_agnostic	M4m	welding_line	AP50 + recall@0.25	1.5e-05		0.0
preprocessing_at_test_time	M1m	gc10: raw vs preprocessed	mAP@0.5	raw=0.044836 pp=0.038581		
preprocessing_at_test_time	M1m	neu: raw vs preprocessed	mAP@0.5	raw=0.734471 pp=0.531359		
preprocessing_at_test_time	M2m	gc10: raw vs preprocessed	mAP@0.5	raw=0.036962 pp=0.028583		
preprocessing_at_test_time	M2m	neu: raw vs preprocessed	mAP@0.5	raw=0.51118 pp=0.737584		
preprocessing_at_test_time	M3m	gc10: raw vs preprocessed	mAP@0.5	raw=0.028274 pp=0.024334		
preprocessing_at_test_time	M3m	neu: raw vs preprocessed	mAP@0.5	raw=0.746039 pp=0.53241		
preprocessing_at_test_time	M4m	gc10: raw vs preprocessed	mAP@0.5	raw=0.033872 pp=0.024474		
preprocessing_at_test_time	M4m	neu: raw vs preprocessed	mAP@0.5	raw=0.512938 pp=0.754711		

Findings. (i) Absolute zero-shot performance is near-zero for every variant (class-agnostic mAP@0.5 $\approx$ 0.03–0.05 on all 2,300 images). (ii) No modification helps transfer: relative to baseline, M4 is -1.03 pp ($p < 0.001$), M3 -1.62 pp ($p < 0.001$), M2 -0.73 pp ($p = 0.014$) on all images; on the 230-image test split all comparisons are null ($|\Delta| \leq 1$ pp, $p \geq 0.40$). The old report's "+49.8% relative" (0.0311 $\rightarrow$ 0.0466 on 230 images) does not survive either the whole-dataset scope or the matched protocol. (iii) Resolution/scale controls: evaluating at imgsz 1024 raises all-images mAP@0.5 for both baseline and full model ($\approx +2.0$ pp and $+0.5$ pp), and defect-scale-matched tiling (640 $\times$ 640 native crops)

lowers

absolute AP further while

preserving the ordering (baseline 0.0244 vs full model 0.0150) — the transfer conclusion is not an artifact of downscaling 2048 $\times$ 1000 to 640. (iv) Per-GC10-class breakdown: what little transfer exists lands on blob/texture-like classes (oil_spot, punching_hole, crescent_gap, water_spot) and is ≈ 0.001 on the semantically closest classes (inclusion, rolled_pit). Cross-dataset "transfer" here reflects superficial texture similarity, not class semantics — which also explains why the restricted-overlap protocol is at noise level for every model (≤ 0.0022 mAP@0.5; original-model table in Section 4, matched-model rows in

results_v3/audit/gc10_restricted_matched.csv). (v) Preprocessing at test time cannot rescue transfer: preprocessed-input models evaluated on preprocessed GC10 remain $\leq$ baseline-on-raw (matched_cross_dataset.csv , *_pp protocols).

5.5 Efficiency

Generators: scripts_v3/fps_bench.py (matched runs: results_v3/audit/matched_fps_benchmark_summary.csv , matched_gpu_state.csv ; Phase 3: p3_fps_*).

variant	FPS fp32	$\pm$ std	FPS fp16	$\pm$ std	inference ms fp32
M1	65.03	0.54	63.17	1.0	10.57
M2	66.29	0.88	65.04	0.32	10.802
M3	59.92	0.82	58.22	0.66	11.495
M4	62.13	0.63	60.59	1.26	11.708

Findings. Under a controlled protocol (batch 1, 5 repeats, warm-up, one session) all variants run at 58–67 FPS fp32 wall-clock on the RTX 3050 Laptop — far above the 30–45 FPS "real-time" band the old report asserted without derivation. Attention costs $\approx$ 2 ms/frame of inference time (+13–14%: baseline $\approx$ 14.1 ms vs attention $\approx$ 16.0 ms in the audit session; SE/CBAM $\approx$ 11.0/11.9 ms in the Phase-3 session — cross-session numbers are not comparable on this WDDM laptop GPU, within-session ones are). fp16 gives no reliable speedup at batch 1. Wall-clock FPS carries session-level uncertainty up to $\pm$ 10–20% and input-content effects up to $\sim$ 35%; only the internal inference-time breakdown is stable. The old table's 43.5–59.5 FPS variant spread is not reproducible and should not be cited.

5.6 Quantified Grad-CAM

Generator: scripts_v3/gradcam_quant.py $\rightarrow$ results_v3/phase3/gradcam_quant.csv .

Whole NEU test split (270 images), matched models M1/M3/M4.

model	scale	PG all	95% CI	PG cond (TP@0.25)	95% CI	energy-in-GT	95% CI
M1m	P3	0.4852	[0.4222, 0.5444]	0.5066	[0.441, 0.5677]	0.4483	[0.4201, 0.4783]
M1m	P4	0.4481	[0.3889, 0.5037]	0.4585	[0.3974, 0.5197]	0.5033	[0.4693, 0.537]
M1m	P5	0.2222	[0.174, 0.2741]	0.2052	[0.1528, 0.262]	0.5762	[0.5122, 0.6379]
M1m	fused	0.4889	[0.4333, 0.5481]	0.4978	[0.4366, 0.5633]	0.4795	[0.4486, 0.5098]
M3m	P3	0.5296	[0.4667, 0.5853]	0.5579	[0.4959, 0.6157]	0.4629	[0.4339, 0.4919]
M3m	P4	0.4333	[0.3741, 0.4889]	0.4339	[0.3719, 0.4959]	0.4787	[0.4454, 0.5117]
M3m	P5	0.237	[0.1889, 0.2926]	0.2273	[0.1776, 0.2851]	0.5058	[0.4347, 0.5804]
M3m	fused	0.4815	[0.4185, 0.5407]	0.4793	[0.4132, 0.5372]	0.4648	[0.4347, 0.4967]
M4m	P3	0.5444	[0.4852, 0.6037]	0.5521	[0.4785, 0.6259]	0.5078	[0.4801, 0.5355]
M4m	P4	0.5333	[0.4778, 0.589]	0.6319	[0.5644, 0.7055]	0.6474	[0.6146, 0.6782]
M4m	P5	0.2963	[0.2444, 0.3556]	0.2577	[0.1963, 0.3313]	0.5103	[0.4589, 0.5617]
M4m	fused	0.5222	[0.463, 0.5778]	0.5337	[0.454, 0.6074]	0.5461	[0.5179, 0.5728]

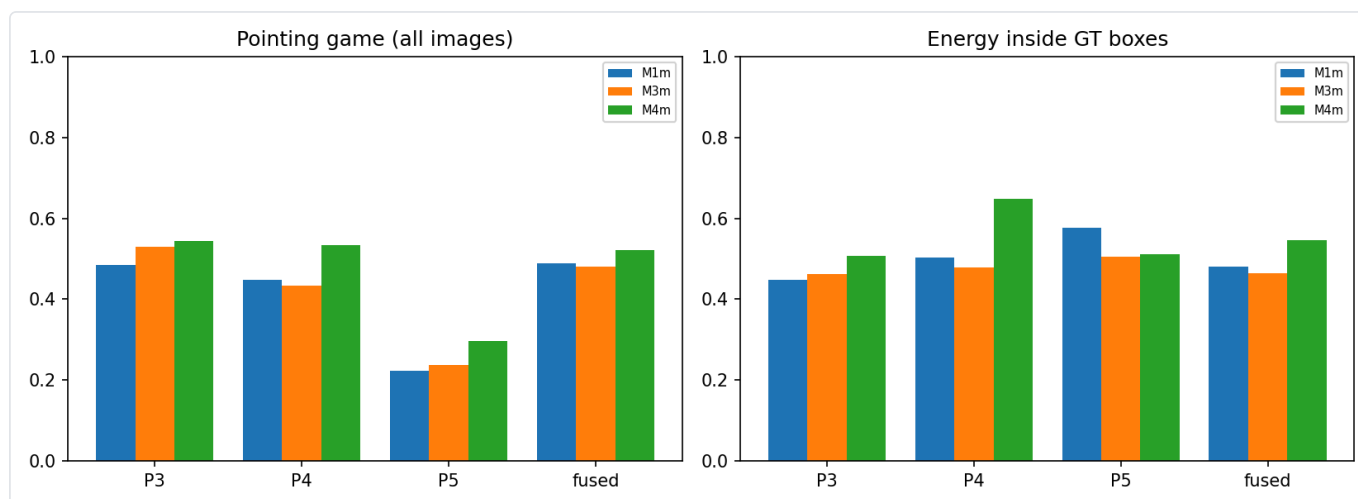


Figure: Pointing game and energy-in-GT for matched models at P3/P4/P5/fused.

Findings. Findings. (i) Localisation quality is scale-dependent: the pointing game is highest at P3 (0.49-0.54 across models) and lowest at P5 (0.22-0.30) - the 20x20 P5 maps used in the old report's qualitative gallery are the coarsest possible evidence, and boundary-level claims from them are not supportable. (ii) Fused-map pointing games overlap broadly across models (M1m 0.489 [0.433, 0.548], M3m 0.482 [0.419, 0.541], M4m 0.522 [0.463, 0.578]): no evidence that attention sharpens peak localisation. (iii) Energy-in-GT is highest for the full model (M4m fused 0.546 [0.518, 0.573]; M4m P4 0.647 [0.615, 0.678]) vs baseline (M1m fused 0.480 [0.449, 0.510]) and attention-only (M3m 0.465 [0.435, 0.497]), with non-overlapping CIs - the full model places a larger share of its attribution mass inside defects even though its peak is not more accurate. (iv) Conditional pointing-game values use different denominators ($n_{\text{cond}} = 229/242/163$ images with a TP at $\text{conf} \geq 0.25$ for M1m/M3m/M4m) and are therefore NOT directly comparable across models - selection confound; the unconditional numbers are the primary evidence.

6. Discussion

Hypothesis verdicts (criteria fixed in advance: "supported" = paired 95% CI excludes 0 in the predicted direction on the primary metric; "inconclusive" = CI spans 0 with point estimate in the predicted direction; "not supported" otherwise):

- **H1 — illumination preprocessing improves in-domain detection: NOT SUPPORTED.** +0.31 pp mAP@0.5 ($p=0.67$), -0.22 pp mAP@0.5:0.95 ($p=0.72$); the grayscale-CLAHE control is worse than both raw and PP-original. On grayscale data the LAB rationale is void, and the round trip injects chroma noise. Preprocessing also **harms** zero-shot transfer (-0.73 pp, $p=0.014$) and creates a hard input-distribution dependency (PP-trained models lose ≈ 20 pp mAP@0.5 on raw test inputs).
- **H2 — lightweight multi-scale attention improves in-domain detection: SUPPORTED (weakly) for mAP@0.5, INCONCLUSIVE for mAP@0.5:0.95, NOT SUPPORTED as module-specific.** Full model +2.02 pp mAP@0.5 ($p=0.038$); attention-only +1.16 pp ($p=0.18$); SE statistically ties or beats ECA+spatial in-domain and dominates it on transfer. The reliable, large effect is class-specific: crazing +8.5...+10.6 pp for every attention variant.
- **H3 — variants remain real-time: SUPPORTED**, with the caveat that the original FPS evidence was unreliable; under a controlled protocol all variants are ≥ 58 FPS fp32 at batch 1 and attention costs $\approx +13\%$ inference latency.
- **H4 — the combination synergises under cross-dataset shift: NOT SUPPORTED.** Zero-shot GC10 performance is near-zero for all variants; every modification is neutral-to-harmful relative

to baseline on all 2,300 images (M4 – 1.03 pp, $p < 0.001$); the old "+49.8%" was a 230-image subset artifact of a confounded protocol.

Why the old study reached opposite conclusions. Three mechanisms, all documented:

(1)

Confounded training

: fine-tuning a converged baseline for 25–30 epochs with a frozen backbone measures fine-tuning dynamics, not the interventions — the identity anchor outscoring all trained variants proves the budget/init, not the modules, drove those numbers, and the per-class "regressions" reversed under the matched protocol. (2)

Small-subset selection

:

230 of 2,300 GC10 images; the sign of the M4–M1 transfer difference flips with scope.

(3)

Reporting errors

: wrong-checkpoint class table, unsourced mAP@0.5:0.95 column, cherry-picked abstract numbers, 309-vs-315 parameter inconsistency.

What survives of "IAMA-Net". A ~300-parameter attention addition at P3/P4/P5 gives a modest, mostly-crazing-driven in-domain mAP@0.5 gain at $\approx 13\%$ inference-latency cost; the illumination preprocessing component should be dropped (no in-domain benefit, transfer harm, input-distribution lock-in); and none of the components improves cross-dataset robustness. For deployment on a new mill's data, retraining on in-domain data remains necessary — no variant meaningfully detects GC10 defect classes zero-shot.

7. Limitations and threats to validity

1. **Single seed per variant** (user-directed for time). Between-seed variance is unknown; NEU mAP@0.5 CI half-widths from image resampling alone are $\approx \pm 3.3$ pp, and the audit showed the original single-seed noise floor at ≈ 1 pp. The +2.02 pp ($p=0.038$) headline is conditional on seed 0; a seed sweep is the first thing to add for a paper submission.
2. **Image-level bootstrap CIs** quantify evaluation-sampling noise only, not training noise.
3. **4 GB laptop GPU (WDDM, shared with display)**: FPS wall-clock is session-dependent; we report within-session comparisons and internal stage timings.
4. **Grad-CAM target choice** (max pre-NMS anchor score) is one of several defensible targets; heatmap quality on 200×200 inputs upscaled to 640 is inherently coarse (P5 maps are 20×20).
5. **Tiling control** uses non-overlapping 640 crops; defects cut by tile boundaries are penalised, which lowers absolute tiled AP for all models equally.
6. **Restricted-overlap mapping** (NEU inclusion/pitted/rolled-in-scale $\leftrightarrow$ GC10 inclusion/rolled_pit) is a judgement call, fixed before evaluation and documented; GC10 test split contains a single rolled_pit instance, so only the all-images scope is informative.
7. **Reconstructed attention class**: the original patched ultralytics class is lost; our reconstruction reproduces all originally reported checkpoint metrics to ≤ 0.0002 , but forward semantics beyond those metrics are inferred.
8. **Old-report numbers**: several (mAP@0.5:0.95 column, class-wise tables) are not reproducible from any artifact in the repository; we treat the recomputed values as the only valid record of the original checkpoints.
9. **GC10 "all images"** includes its original train/val splits. This is legitimate for a zero-shot evaluation (no GC10 data was used for training or model selection), but numbers are not comparable to supervised GC10 results in the literature.

8. Conclusion

Under a matched protocol with pre-registered statistics, a 315-parameter multi-scale attention addition to YOLO11s yields a small in-domain mAP@0.5 gain (+2.0 pp, $p=0.038$, single seed) concentrated in the hardest class (crazing $\approx +10$ pp), at $\approx 13\%$ inference-latency cost, with mAP@0.5:0.95 inconclusive and no module-specific advantage over SE. Illumination preprocessing provides no measurable in-domain benefit and measurably harms zero-shot transfer. Zero-shot cross-dataset detection on GC10-DET is near-zero for all variants, and the previously reported transfer "synergy" is a protocol artifact. The study's durable contributions are methodological: a reproducible audit trail from flawed claims to corrected evidence, a matched-protocol ablation with uncertainty quantification, and a whole-dataset zero-shot transfer protocol with explicit scale/label-space/input-distribution controls.

References

Verified status per reference is recorded in `results_v3/report/references.md` (arXiv/Crossref checks, 2026-10-05). Items marked $\triangle$ require one manual confirmation before external submission.

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7. C. Tomasi, R. Manduchi. "Bilateral Filtering for Gray and Color Images." ICCV 1998. DOI 10.1109/ICCV.1998.710815.
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9. S. M. Yasir, H. Ahn. "Faster Metallic Surface Defect Detection Using Deep Learning with Channel Shuffling." Computers, Materials & Continua, 2023. arXiv:2406.14582, DOI 10.32604/cmc.2023.035698.
10. Ultralytics. "YOLO (v8/v11) documentation." <https://docs.ultralytics.com> — version 8.3.40 used throughout this study.
11. $\triangle$ GC10-DET dataset paper, commonly cited as: X. Lv et al., "GC10-DET: An Industrial Surface Defect Detection Dataset Using Two-Stage Cascade Network," IEEE Trans. Industrial Informatics, 2023 — bibliographic details could not be verified from this environment (Crossref/OpenAlex unavailable); verify manually or cite the dataset release + PeerJ CS 2024, DOI 10.7717/peerjcs.1727 (verified) which documents the benchmark.
12. $\triangle$ K. Zuiderveld. "Contrast Limited Adaptive Histogram Equalization." Graphics Gems IV, Academic Press, 1994, pp. 474–485 — canonical chapter; DOI string to be re-verified (commonly cited as 10.1016/B978-0-12-336156-1.50061-7).

Appendix A — Artifact and script index

artifact	generator
<code>results_v3/audit/audit_report.md</code> (+ all audit CSVs)	<code>scripts_v3/audit_data.py</code> , <code>count_params.py</code> , <code>identify_preprocessing_params.py</code> , <code>eval_unified.py</code> , <code>bootstrap_ci.py</code> , <code>gc10_restricted_overlap.py</code> , <code>fps_bench.py</code> , <code>val_batch_sensitivity.py</code> , <code>reconciliation.py</code> , <code>make_audit_report.py</code>
<code>results_v3/ablation_matched.csv</code> , <code>cross_dataset_matched.csv</code>	<code>run_phase2_eval.py</code> → <code>eval_unified.py</code> , <code>bootstrap_ci.py</code> , <code>gc10_restricted_overlap.py</code> , <code>fps_bench.py</code> , <code>aggregate_matched.py</code>
<code>results_v3/attention_alternatives.csv</code> , <code>preprocessing_controls.csv</code> , <code>transfer_controls.csv</code>	<code>run_phase3_eval.py</code> → same + <code>transfer_controls.py</code> , <code>aggregate_phase3.py</code>
<code>results_v3/phase3/gradcam_quant.csv</code> , <code>transfer_*.csv</code>	<code>gradcam_quant.py</code> , <code>transfer_controls.py</code>
<code>results_v3/report/tables/*.md</code> , <code>figures/*.png</code>	<code>make_report_tables.py</code>
training runs + manifests	<code>train_matched.py</code> via <code>run_queue.py</code> (<code>results_v3/queue_phase2.json</code> , <code>queue_phase3.json</code>); mapped inits via <code>map_pretrained.py</code>
this document	<code>build_report.py</code> from <code>report/report_template.md</code>